

What a stellar occultation forces about a body's size when only its silhouette's convexity is assumed

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September 2026

Abstract

A stellar occultation gives a handful of parallel chords across a small body's silhouette, and the size that gets published is an ellipse fitted to them. We ask what the chords force on their own. Because occultation chords are parallel and a convex silhouette has a concave upper boundary and a convex lower one, the chord length $w(y)$ is a *concave* function of the cross-track coordinate, vanishing outside the body's support; the silhouette's area is $\int w$. The problem is therefore one-dimensional and closed form, with no optimiser and no convergence, and on published decimal data every bound is an exact rational. Applied to the 2017 May 20 occultation by the centaur (95626) 2002 GZ₃₂, convexity alone forces the area-equivalent diameter into $[168.3, 267.5]$ km, against a published 206 ± 15 km. Two facts fall out. At face value the five chords admit *no* convex silhouette at all — the concave hull of the measured lengths runs 5.5 km above one station's own chord — and about half of one stated error bar is needed before one exists. And the bound is asymmetric by construction: the floor is bought by the chords, while the ceiling is bought entirely by the telescopes that recorded *nothing*. Moving this campaign's 23 negative stations out of reach raises the ceiling from 267.5 to 577.0 km.

1 The chord-length function

Occultation chords are parallel: the shadow sweeps in one direction and every station cuts the silhouette along that direction. Take that direction as x and let y be the cross-track coordinate. Let $K \subset \mathbb{R}^2$ be the silhouette, assumed compact and convex, with slices $K \cap \{y\} = [L(y), R(y)]$ on its support $[Y^-, Y^+]$.

Proposition 1. *R is concave and L is convex on $[Y^-, Y^+]$, so the chord length $w = R - L$ is concave there and zero outside; and $\text{area}(K) = \int w$.*

Proof. For y_1, y_2 in the support and $t \in [0, 1]$, the points $(R(y_1), y_1)$ and $(R(y_2), y_2)$ lie in K , so their combination does too, giving $R(ty_1 + (1-t)y_2) \geq tR(y_1) + (1-t)R(y_2)$; the same argument with L reverses the inequality. The area statement is Fubini. $\square$

Remark. *Convexity is weaker than the ellipse that occultation papers fit, and it is already asserted by the data: a single ingress and a single egress per station is what a convex silhouette produces. It is also capable of being wrong, which is what makes it an assumption worth naming — a dumbbell has a chord-length function that is not concave, and the construction below excludes it.*

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Write the measurements as $w(y_i) \in [\ell_i - \varepsilon_i, \ell_i + \varepsilon_i]$ at n cross-track positions, and let $y^- < \min_i y_i$ and $y^+ > \max_i y_i$ be the nearest stations that observed and saw nothing, so that $[Y^-, Y^+] \subseteq [y^-, y^+]$.

2 The two bounds, and why they are asymmetric

Proposition 2 (floor). *Let $\hat{w}$ be the concave hull of the points $(y_i, \ell_i - \varepsilon_i)$ on $[y_1, y_n]$. Then $\text{area}(K) \geq \int_{y_1}^{y_n} \hat{w}$, and the bound is attained.*

Proof. w is concave on a support containing $[y_1, y_n]$ and satisfies $w(y_i) \geq \ell_i - \varepsilon_i$, so it dominates the smallest concave function through those points; and $w \geq 0$ elsewhere. Attainment: the convex hull of the chords themselves is a convex body realising it. $\square$

Remark. *The negative stations do not appear. Nothing forces the body past its outermost chord: it may end flush with it, giving a flat edge, which is a perfectly good convex body. In particular the polygon obtained by “joining the dots” down to the negative stations is not a lower bound — it already assumes the body reaches them.*

Proposition 3 (ceiling). *For $y \notin [y_i, y_j]$ with $y_i < y_j$,*

$$w(y) \leq U_{ij}(y) := \begin{cases} (\ell_j + \varepsilon_j) + \frac{(\ell_j + \varepsilon_j) - (\ell_i - \varepsilon_i)}{y_j - y_i}(y - y_j), & y > y_j, \\ (\ell_i + \varepsilon_i) + \frac{(\ell_i + \varepsilon_i) - (\ell_j - \varepsilon_j)}{y_i - y_j}(y - y_i), & y < y_i, \end{cases} \quad (1)$$

and consequently $\text{area}(K) \leq \int_{y^-}^{y^+} \max(0, \min_{ij} U_{ij})$.

Proof. For $y_i < y_j < y$, concavity of w on the support gives $w(y_j) \geq \frac{(y - y_j)w(y_i) + (y_j - y_i)w(y)}{y - y_i}$; solving for $w(y)$ gives a bound in which $w(y_j)$ carries a positive coefficient and $w(y_i)$ a negative one, so the largest admissible value uses $\ell_j + \varepsilon_j$ and $\ell_i - \varepsilon_i$. The other case is symmetric. Outside $[y^-, y^+]$ the body is absent, so $w = 0$ there. $\square$

Remark (the negatives are not samples). *It is tempting to enter $y^\pm$ into the pair list as ordinary samples with $w = 0$. That is unsound. w is concave on its support and zero outside; a function that is zero on two rays and concave between them is not concave. On a triangular silhouette, admitting the far cap as a sample produced a ceiling of 7529 against a true area of 9000, because the cap dragged the envelope down through the body itself. The negative stations clip the domain of integration and nothing else — and that is already the whole of what they buy.*

Remark (the ceiling is discontinuous). *The constraints anchored at a sample are valid only up to it, so $\min_{ij} U_{ij}$ jumps upward the instant y passes a chord. Integrating it as a polyline through the sample points understates the area — by 16% on a circle — and makes the ceiling fall when chords are removed, which is the wrong direction and is how the error was found. Between breakpoints the envelope is affine, so the midpoint rule is exact and blind to the jumps.*

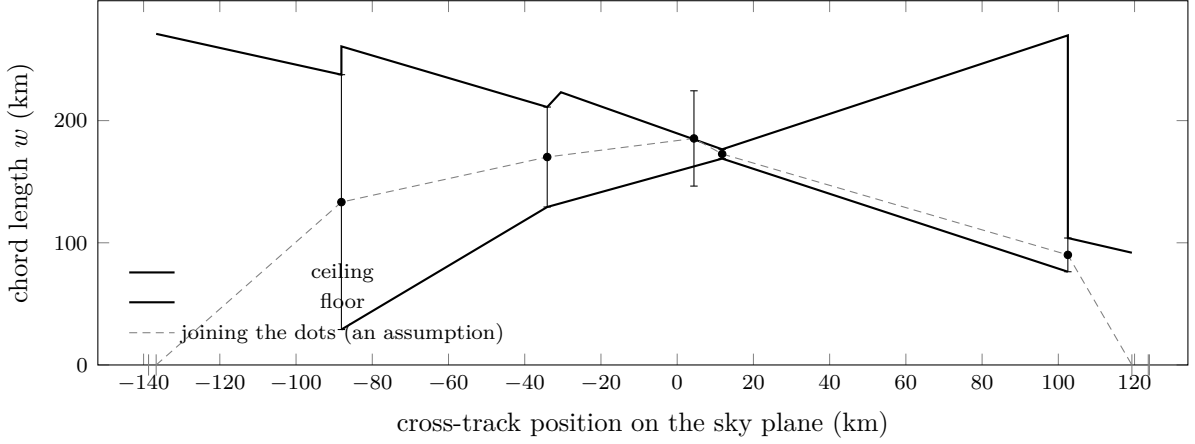


Figure 1: The five chords with their published errors, the 23 stations that saw nothing (ticks at zero), and the certified envelopes. Joining the dots is drawn dashed because it is an assumption, not a bound.

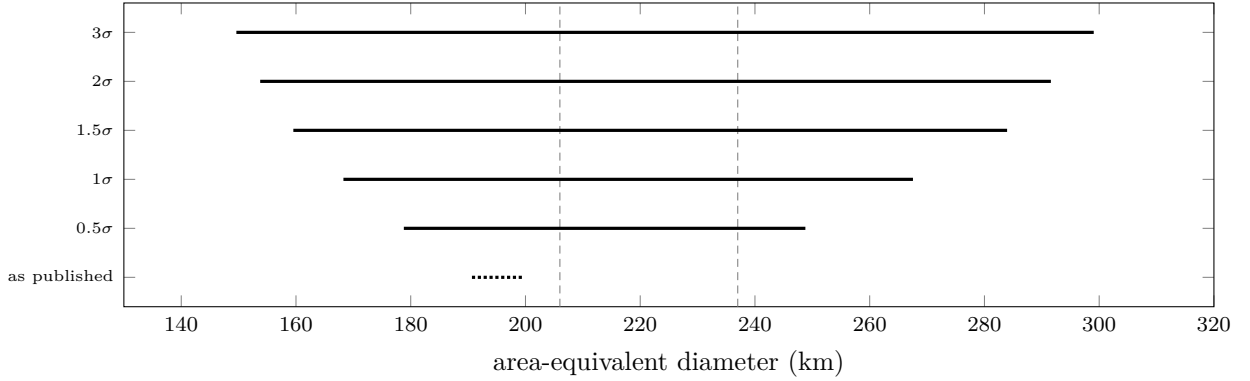


Figure 2: Certified area-equivalent diameter against the error budget. The top bar is dotted because at face value no convex silhouette fits the chords. Vertical lines: the published ellipse 206 km and the radiometric 237 km.

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The campaign of Santos-Sanz et al. (2021) is used because its site table publishes, for every station, the projected distance on the sky plane to the reconstructed centerline in kilometres together with the detection status. That column is the cross-track coordinate, already reduced, so no ephemeris, station projection or Earth-rotation model enters here. Five chords were positive and 23 stations saw nothing. A timing shift slides a chord along its own line and does not change its length, so nothing below depends on which of the paper’s two fits — “original” ($\chi^2 = 29.5$) or “shifted” ($\chi^2 = 4.0$) — one prefers.

At the published 1σ chord errors, convexity alone gives

$$D_{\text{eq}} \in [168.3, 267.5] \text{ km},$$

against a published 206 ± 15 km. Two consequences are worth stating.

The chords are not consistent with convexity as published. At zero error the concave hull of the measured lengths runs 5.5 km above the Javalambre chord; about half of one stated error bar admits a convex body. The published analysis meets the same fact from the other side and records it as $\chi^2 = 29.5$.

A disagreement in the literature dissolves. The paper reports its occultation diameter and notes that its mean three-dimensional estimate is not in agreement with the radiometric 237 ± 8 km from Herschel, Spitzer and ALMA. Both values lie inside the certified interval. The tension is between two models, not between two measurements.

What the misses are worth. Moving the 23 negative stations out of reach raises the ceiling from 267.5 to 577.0 km. The telescopes that recorded nothing are worth 309 km of upper bound — more than the body’s own diameter — and they are the part of a campaign least often published.

4 Limitations

Only one event is analysed, because a second requires a campaign that publishes its cross-track distances; converting a timings-only campaign means redoing the station projection and Earth rotation, which has a built-in check (the computed chord lengths must reproduce the published ones) but is a real piece of geometry with a real chance of a silent error. The position of each chord along its own line is not used — it is not published here, and the area bound does not need it — but it would tighten the ceiling, because it constrains L and R separately rather than only their difference. A central-symmetry rung, which would make w symmetric about the centre, is a genuine further constraint and a genuine test; these bodies are not symmetric, and a refutation would be the informative outcome.